NAME: - SREEJA SANYAL

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NEAR-FINAL PROJECT REPORT

1. Introduction

Lithium-ion batteries are widely used in electric vehicles, portable electronics, and energy storage systems. Knowing the battery's State of Charge (SOC) and estimating the remaining charging time can help users understand the battery's current condition and how much longer charging may take.

The aim of this project is to develop machine learning models that estimate battery SOC and predict the Remaining Charging Time using battery measurements and derived features.

2. Project Objectives

- Estimate the State of Charge of a lithium-ion battery.

- Predict the Remaining Charging Time.

- Study battery measurements such as voltage, current, temperature, and capacity.

- Compare machine learning models using evaluation metrics.

- Organize the final models, notebooks, and supporting outputs for reproducibility.

3. Methodology

1. Dataset and Preprocessing:-

   The project uses lithium-ion battery charging data containing measurements such as voltage, current, temperature, capacity, and accumulated energy.The dataset was processed and prepared for model training and evaluation. Relevant features were selected, and additional features such as elapsed time and power were used in the modelling workflow.

2. Feature Selection

The final model workflow uses the following features:

- Voltage

- Current

- Temperature

- Elapsed Time

- Cycle

- Power

These features provide information about battery operating conditions and charging behavior.

3. Model Selection:- Two machine learning models were selected for the project:

- XGBoost Regressor: Used for State of Charge estimation.

- Random Forest Regressor: Used for Remaining Charging Time prediction.

The models were selected based on the model-comparison and evaluation work carried out during the project.

4. Model Evaluation

Model performance was evaluated using:

- MAE (Mean Absolute Error): Measures the average absolute difference between predicted and actual values.

- RMSE (Root Mean Squared Error): Gives greater weight to larger prediction errors.

- R^2 (Coefficient of Determination): Indicates how well the model explains variation in the target values.Validation and test results was reviewed to assess model performance.

4. Results and Discussion

The selected models for the two prediction tasks are XGBoost Regressor for SOC estimation and Random Forest Regressor for Remaining Charging Time prediction.

The previously recorded test results are:

- SOC estimation — XGBoost Regressor: Test $R^2 = 0.990521$.

- Remaining Charging Time — Random Forest Regressor: Test $R^2 = 0.998756$.

The Remaining Charging Time model also recorded a validation R^2 of 0.996502 in the saved project results. These scores indicate strong performance on the evaluated data. However, the scores alone do not guarantee equally accurate predictions for every battery or operating condition.The final numerical results should be checked against the corresponding saved evaluation notebooks before submission.

5. Software Artefacts

The project materials include the following artefacts, where available:

- Google Colab notebooks used for data processing, training, and evaluation.
- Lithium-ion battery dataset and processed data.
- Saved XGBoost model for SOC estimation.
- Saved Random Forest model for Remaining Charging Time prediction.
- Evaluation tables and supporting plots.

- Project source files and documentation.
- Relevant model output files.

These artefacts support the review and continuation of the project workflow.

6. Reproducibility

The project workflow was organised so that the main steps could be followed from the notebooks and supporting files. These steps include data preprocessing, feature selection, model training, and model evaluation. For reproducibility, the dataset location, required Python libraries, notebook sequence, and saved model paths should be checked. Any missing dependency or file path should be documented and corrected before final submission.

7. Limitations

- Prediction errors may occur under certain charging conditions.
- Model performance may vary across battery temperatures and operating conditions.
- The quality and coverage of the dataset can affect model performance.
- Further evaluation using additional battery datasets may help assess generalisation to other operating conditions.

8. Conclusion

The project developed machine learning models for lithium-ion battery SOC estimation and Remaining Charging Time prediction. XGBoost Regressor was selected for SOC estimation, while Random Forest Regressor was selected for Remaining Charging Time prediction.The project notebooks, evaluation outputs, and saved models form the main technical artefacts. The near-final report and presentation have been prepared for review, with final corrections and approval to follow the required internship process.